

TraceFund

Blockchain-Based Transparent Fund Disbursement and Fraud Detection System

Group No: 19

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1. Requirement Collection

Problem Statement

Public funds such as subsidies, disaster relief, and welfare payments are disbursed as cash or standard bank transfers, with no mechanism to control or verify how the money is ultimately spent. This contributes to fund leakage and misuse. The system will issue funds as digital tokens with spending restrictions enforced through smart contracts, record every transaction immutably on a blockchain, and flag suspicious transaction patterns.

Functional Requirements

- Government must be able to issue tokens tied to a specific fund type and spending category
- Beneficiaries must be able to receive and spend tokens only at approved vendors
- Vendors must be able to register and get approved for specific spending categories
- Transactions violating spending rules must be rejected automatically
- All transactions must be recorded on the blockchain
- Suspicious transaction patterns must be flagged for auditor review
- Public dashboard must show aggregate fund utilization without exposing individual data

Non-Functional Requirements

- Transaction validation should complete within a few seconds for a usable demo experience
- System should be built using Python (FastAPI) for backend and React/Next.js for frontend
- Data privacy: individual transaction details must not be exposed on the public dashboard

Stakeholders

- Government / Treasury department (fund issuer)
- Beneficiaries (citizens receiving funds)
- Vendors (businesses accepting restricted funds)
- Auditors / Regulators
- General public (transparency viewers)

2. Initial User Stories

ID	Role	User Story
US-01	Government Admin	As a government admin, I want to issue fund tokens with defined spending categories so that money is restricted to its intended purpose.
US-02	Government Admin	As a government admin, I want to view a dashboard of all disbursed funds so that I can monitor overall usage.

US-03	Beneficiary	As a beneficiary, I want to receive fund tokens in a digital wallet so that I can access my allocated funds.
US-04	Beneficiary	As a beneficiary, I want to spend my tokens only at approved vendors so that my transaction goes through without manual approval delays.
US-05	Vendor	As a vendor, I want to register my business under specific categories so that I can accept restricted-category tokens.
US-06	Vendor	As a vendor, I want to view incoming payments so that I can confirm successful transactions.
US-07	Auditor	As an auditor, I want to view flagged suspicious transactions so that I can investigate possible misuse.
US-08	Auditor	As an auditor, I want to access the full transaction trail for any fund so that I can verify compliance.
US-09	Public User	As a member of the public, I want to view aggregate, category-wise fund utilization so that I can see how public funds are being used.
US-10	System	As the system, I want to automatically reject a transaction that violates spending rules so that misuse is prevented before it happens.

3. Modules and Ownership

Module	Description	Assigned To
Smart Contract & Token Issuance	Solidity contracts handling token creation and spending-rule logic	Johan
Blockchain Integration Layer	Web3.py bridge connecting backend to deployed smart contracts	Johan
Rules Engine	Validates each transaction against fund category and vendor approval	Johan
Fraud Detection	Graph-based analysis of transaction patterns to flag suspicious activity	Johan
Database & Backend API	Core data models, all REST endpoints connecting every module	Johan
Wallet Management UI	Beneficiary-facing screens: balance display, transaction history, spend action	Srilakshmi
Government Admin Dashboard UI	Fund issuance form and disbursement monitoring view	Srilakshmi
Public Transparency Dashboard UI	Aggregate, category-wise fund utilization view	Srilakshmi
Vendor Registration UI	Vendor sign-up form and category selection screen	Abel
Vendor Approval UI	Admin screen to approve/reject pending vendor registrations	Abel
Auditor Dashboard UI	Flagged transaction list and review-status screen	Abel

4. Initial Database Schema

Total tables: 7 (Users, Wallets, FundTypes, Vendors, Transactions, FraudFlags, AuditLogs)

Table: Users

Stores all system users across roles (Government, Beneficiary, Vendor, Auditor)

Column	Type	Description
id	INTEGER (PK)	Unique user identifier
name	VARCHAR	Full name
email	VARCHAR	Login email
password_hash	VARCHAR	Hashed password
role	VARCHAR	Government / Beneficiary / Vendor / Auditor
created_at	TIMESTAMP	Account creation date

Table: Wallets

Digital wallet holding a user's token balance

Column	Type	Description
id	INTEGER (PK)	Unique wallet identifier
user_id	INTEGER (FK)	References Users.id
balance	DECIMAL	Current token balance
wallet_address	VARCHAR	Blockchain wallet address

Table: FundTypes

Defines a category of public fund and its spending restrictions

Column	Type	Description
id	INTEGER (PK)	Unique fund type identifier
name	VARCHAR	e.g., Disaster Relief, Education Subsidy
allowed_category	VARCHAR	Permitted vendor category for this fund
description	TEXT	Purpose of the fund

Table: Vendors

Registered vendors approved to accept restricted-category tokens

Column	Type	Description
id	INTEGER (PK)	Unique vendor identifier

user_id	INTEGER (FK)	References Users.id
business_name	VARCHAR	Registered business name
category	VARCHAR	Vendor's registered category (food, medical, etc.)
approval_status	VARCHAR	Pending / Approved / Rejected

Table: Transactions

Record of every fund transfer between wallets

Column	Type	Description
id	INTEGER (PK)	Unique transaction identifier
sender_wallet_id	INTEGER (FK)	References Wallets.id
receiver_wallet_id	INTEGER (FK)	References Wallets.id
fund_type_id	INTEGER (FK)	References FundTypes.id
amount	DECIMAL	Transaction amount
status	VARCHAR	Approved / Rejected
blockchain_tx_hash	VARCHAR	On-chain transaction reference
timestamp	TIMESTAMP	Time of transaction

Table: FraudFlags

Suspicious transactions flagged by the fraud detection module

Column	Type	Description
id	INTEGER (PK)	Unique flag identifier
transaction_id	INTEGER (FK)	References Transactions.id
reason	VARCHAR	Why it was flagged
reviewed_by	INTEGER (FK)	References Users.id (auditor)
review_status	VARCHAR	Pending / Cleared / Confirmed Misuse
created_at	TIMESTAMP	When the flag was raised

Table: AuditLogs

System-wide action log for accountability

Column	Type	Description
id	INTEGER (PK)	Unique log entry identifier
actor_id	INTEGER (FK)	References Users.id

action	VARCHAR	Action performed (e.g., fund issued, vendor approved)
target_id	INTEGER	ID of the record affected
timestamp	TIMESTAMP	When the action occurred

[TraceFund Logo]

WalletHistoryProfileLogout

Current Wallet Balance

₹ [amount]

Fund Type: [Disaster Relief]

Spend Funds

[Vendor Name Dropdown]

[Amount Input Field]

Spend

Recent Transactions

Date	Vendor	Category	Amount	Status
[date]	[vendor name]	[Food]	[amt]	[Approved]
[date]	[vendor name]	[Medicine]	[amt]	[Approved]
[date]	[vendor name]	[Other]	[amt]	[Rejected]

[TraceFund Logo]

Issue FundsDisbursementsFlaggedLogout

Issue New Fund

[Select Beneficiary]

[Select Fund Type]

[Amount]

Issue

Disbursement Overview

Total Issued
[₹ amount]

Total Spent
[₹ amount]

Active Beneficiaries
[count]

Recent Disbursements

Date	Beneficiary	Fund Type	Amount
[date]	[name]	[Disaster Relief]	[amt]
[date]	[name]	[Education Subsidy]	[amt]